TEAM FLASHINE

Case Study 3: AI for Cross-Culture Learning

SEAPHERE

REPORT

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1 Introduction

1.1 Background of Case Study

ASEAN is a region rich in cultural diversity, yet meaningful cultural exchange among students across member countries remains limited. While educational systems often emphasize national history and identity, less attention is given to developing a deeper understanding of shared regional values and the everyday experiences of neighboring societies (ASEAN Secretariat, 2016). As a result, gaps in cultural understanding can arise. For instance, behaviors such as silence may be misinterpreted as agreement, or constructive feedback may be perceived as personal criticism (Meyer, 2014). These misunderstandings can hinder effective communication and slow the development of a stronger shared regional identity.

Seaphere seeks to address this challenge by acting as a sovereign Artificial Intelligence (AI) bridge that supports people-centered regional innovation. Powered by SeaphereAI, a Large Language Model (LLM) fine-tuned on ASEAN cultural datasets, the platform goes beyond simple translation by interpreting social cues, guiding culturally appropriate communication, and supporting the development of Intercultural Communicative Competence (ICC). Designed as both an academic and social platform, Seaphere provides students with practical tools to navigate cross-cultural interactions, while also helping prepare a more collaborative and culturally aware workforce for the ASEAN Economic Community (AEC).

1.2 Chosen Sustainable Development Goals (SDG)

Seaphere aligns with the following Sustainable Development Goals to promote regional integration:

- **SDG 4: Quality Education** : Providing structured resources to decode cultural nuances and fostering inclusive digital learning environments.

- **SDG 10: Reduced Inequalities** : Bridging communication gaps for deaf, non-literate, and hearing students through multilingual and sign-language support (UNICEF, 2025).
- **SDG 9: Industry, Innovation, and Infrastructure** : Advancing responsible AI governance and regional innovation partnerships within the ASEAN digital ecosystem (ASEAN Secretariat, 2025).

1.3 Problem Statement

The main challenge lies in the lack of structured and inclusive communication tools capable of interpreting the complex linguistic and social nuances present across Southeast Asia. Although various digital platforms exist to facilitate communication, many of them do not adequately consider the needs of diverse user groups. In particular, students with hearing impairments or individuals from non-literate communities are often overlooked, which can lead to social isolation and reduced participation in collaborative environments (UNICEF, 2025).

In addition, many existing translation technologies focus primarily on converting language from one form to another without accounting for cultural context. As a result, they often fail to capture the social meanings and communication styles embedded within different cultures (MEYER, 2014). This limitation can lead to misunderstandings and increase the risk of intercultural conflict, particularly in academic and professional settings where effective communication is essential.

1.4 Objectives

To solve these issues, Seaphere pursues the following specific goals:

- **Bridge Language Barriers:** Implement real-time translation for text, voice, and sign language to ensure every student's voice is heard.

- **Foster Cultural Empathy:** Deploy a cultural adaptation engine that uses emotion recognition to suggest respectful and context-aware response styles.
- **Enhance Collaboration:** Provide an AI-powered "Collaboration Mode" that translates live discussions and generates summaries to facilitate academic teamwork.
- **Cultivate Regional Identity:** Build a "Community Hub" where AI-generated cultural insights and moderated discussions reduce fragmentation and build a sense of ASEAN belonging

2 Review of Existing Solutions

Most of the existing platforms provide communication and translation support and focus only on basic language conversion without fully addressing and focusing on the user experience of culture, emotions, and accessibility in a multicultural environment like ASEAN. In this section, we discuss many of the existing responses and limitations and introduce Seaphere which will be part of the new solution.

2.1 Google Translate

Google Translate is one example of one of the most used existing translation platforms. Google Translate allows you to translate text, speech, images, and documents across multiple languages to enable users to communicate across languages. However, this platform primarily provides translation without consideration to cross cultural context. This could lead to consequences where some of the translations may sound too direct or inappropriate in some contexts.

2.2 Microsoft Teams / Zoom

Other widely used platforms include Microsoft Teams and Zoom for video conferencing and collaboration, which enable remote communication for the user. These platforms frequently

installed automatic subtitles, and limited support for live translation. However, the platforms are generic meeting tools and not support intercultural communication tools. Microsoft Teams and Zoom cannot offer features that facilitate cultural adaptation suggestions, emotional analysis and sign language interpretation. As a result, these features are less suitable for integration in cross platform academic collaboration.

2.3 Duolingo

Lastly, the existing solution is Duolingo, a language learning platform. Duolingo teaches people new languages using structured classes and exercises. These can be very convenient for a long time learning period but they are not really suitable for real time communication. Moreover, Duolingo lacks on emotion or cultural language cues that are related to emotional or cultural based. Duolingo also does not offer collaboration and multilingual group discussion support, which is very essential for ASEAN students.

2.4 Identified Gaps in Reviewed Existing Solutions

Based on the review of several existing solutions, there are several important limitations have been identified. Several translation existing platforms primarily focus on converting words from one language to another and do not fully understand the cultural meaning behind the message. This causes most of the translated sentences to be too direct, impolite, or inappropriate in certain cultures. In addition, the majority of collaboration systems do not provide multilingual real time translation capabilities. Users have to use a specific single common language when talking with each other, so talking can be complicated for participants that do not speak it fluently which could lead to a reduction in the effectiveness of teamworks, especially among students from different countries.

Moreover, sign language translation systems are typically a standalone application and do not integrate with other communication systems such as chat systems and video meetings. This could

make it hard for people with hearing impairment to join in on group discussions. Finally, most existing translation platforms do not include emotional detection or tone suggestion features. Without these features, the user's intention could be misread, which can accidentally be interpreted to be disrespectful, or rude in some cultures.

In order to overcome all the limitations mentioned above, our proposed system, Seaphere aims to offer an integrated AI powered platform that incorporates features of real time multilingual communication, cultural adaptation and emotion aware interactions, and inclusive collaboration. Additionally, Seaphere also includes new exciting support features for sign language and AI generated summaries of meetings, ensuring effective communication between diverse users.

3 Methodology

3.1 Emphatize

This project aims to enhance cross-cultural communication between deaf, non-literate, and hearing students across ASEAN countries. Based on our observations, individuals with hearing impairments often lack an inclusive platform that allows them to actively participate in group discussions and online forums beyond text-based communication (Alhulays, 2024). This limitation can lead to feelings of disconnection, isolation, and exclusion within digital collaborative environments. For instance, deaf students who are unable to directly listen to their teammates's verbal discussions may struggle to express their ideas effectively. Although they attempt to contribute through text messages, their responses are sometimes overlooked or received delayed attention. Consequently, they may be unfairly perceived as disengaged, less competent, or minimally contributing to the group's progress.

3.2 Define

University students require a more expressive and accessible medium to communicate using sign language, as most existing tools are external applications and are not seamlessly integrated into

learning platforms (Syafira & Yuana, 2026). Furthermore, many of these tools are not culturally adapted to reflect the linguistic diversity and nuances of Southeast Asia. Addressing this issue is essential to ensure equitable communication, collaboration, and interaction without language barriers. The absence of an inclusive communication medium not only complicates academic collaboration but also affects daily interactions, particularly among non-literate communities in Southeast Asia. Therefore, developing a culturally sensitive and accessible solution is crucial in fostering inclusive education and social connectivity within the region.

3.3 Ideate

The team convened in an online meeting to conduct an open brainstorming session. During this session, each member contributed ideas and built upon one another's suggestions to explore the essential features and overall direction of the application. This collaborative exchange of perspectives enabled the team to critically evaluate different possibilities and refine the project's objectives. Each viewpoint played a significant role in shaping the conceptual framework of the solution and its potential contribution to ASEAN communities. Prior to finalizing the project scope, the team considered several possible target user groups. These included the general deaf community and non-literate adults. After careful discussion and evaluation, we decided to focus on Southeast Asian university students, as this demographic is particularly affected by the lack of inclusive communication platforms. By narrowing our target audience, we were able to design a more context-specific and impactful solution tailored to their educational and social needs.

3.4 Prototype

This project employs a high-fidelity prototype developed using Figma as the primary design tool. All essential screens, including account authentication, main feature interfaces, and navigation elements, have been fully designed to reflect the core functionality of the application. The user interface (UI) emphasizes high usability and accessibility, with particular consideration for hearing-impaired users. A minimal colour palette is applied to ensure visual simplicity, clarity, and

consistency across the application. Additionally, the prototype incorporates intuitive layout structures, clear iconography, and easily navigable workflows to enhance the overall user experience (UX). Interactive elements such as buttons, menus, and input fields are designed to provide immediate feedback, making the prototype feel closer to a fully functional application.

3.5 Test

The prototype was subsequently tested by ASEAN students reachable within the team's network. The Figma prototype and corresponding instructions were shared with the participants prior to testing. The instructions outlined the objectives and tasks that participants were expected to complete while interacting with the application. After completing the tasks, participants were asked to fill out a feedback form capturing their demographic information as well as their experience, usability impressions, and suggestions for improvement. Out of 50 participants, the majority provided positive feedback regarding usability, clarity, and feature relevance. Nevertheless, several areas for improvement were identified, including refining certain navigation flows, enhancing feedback for input errors, and providing additional cultural adaptations for sign language gestures. These adjustments are incorporated into the prototype to better meet user needs and ensure a more inclusive and seamless experience.

4 Results

4.1 Helping People Understand Each Other

The Seaphere project successfully created a way for students in ASEAN to connect. Often, we have "cultural misunderstandings" where we misinterpret silence or a polite "yes." Seaphere's prototype shows that AI can do more than just change words from one language to another; it can help us understand the feelings and manners behind those words. This helps students from different countries work together without getting offended or confused.

4.2 Respecting Every Local Voice

ASEAN is home to more than 1,000 languages (Simons & Fennig, 2023). For example, Indonesia has over 700 languages, and the Philippines has over 180 (Ethnologue, 2023). Many AI tools only focus on Western ways of speaking, but Seaphere is different. It is designed to respect our local dialects and ways of talking. This ensures that when a student uses the app, they don't have to lose their own culture to be understood by their neighbors.

Table 1: How Seaphere Features Help ASEAN Students

Feature	**How it helps**	**Why it matters for ASEAN**
Real-time Translation	Allow us to talk easily in more than 10 languages.	Connect the whole region.
Cultural Nuance Tool	Explains local manners and “Saving Face”.	Stop small fights and confusion.
Sign-to-Text Mode	Helps deaf and mute students join the conversation.	Includes everyone in the future.
Community Hub	A place to share stories and traditions.	Makes us unite and understand each other

4.3 Learning the "Unspoken" Rules

In many of our cultures, what we don't say is just as important as what we do say. This is called "high-context" communication (Hall, 1976). For example, a student might stay silent out of respect, but a teammate might think they are ignoring the work. Seaphere’s Nuance Tooltips help by explaining these social rules in real-time. This builds empathy and makes our friendships stronger across borders.

4.4 Including Everyone: Sign Language Support

We want an ASEAN where nobody is left behind. There are about 15 million people in our region who have hearing loss (World Health Organization, 2021). Seaphere includes a special feature that turns Sign Language into text during video calls. This means a deaf student can use their hands to speak, and their teammate can read it instantly. This brings dignity and equal chances to every student in our region.

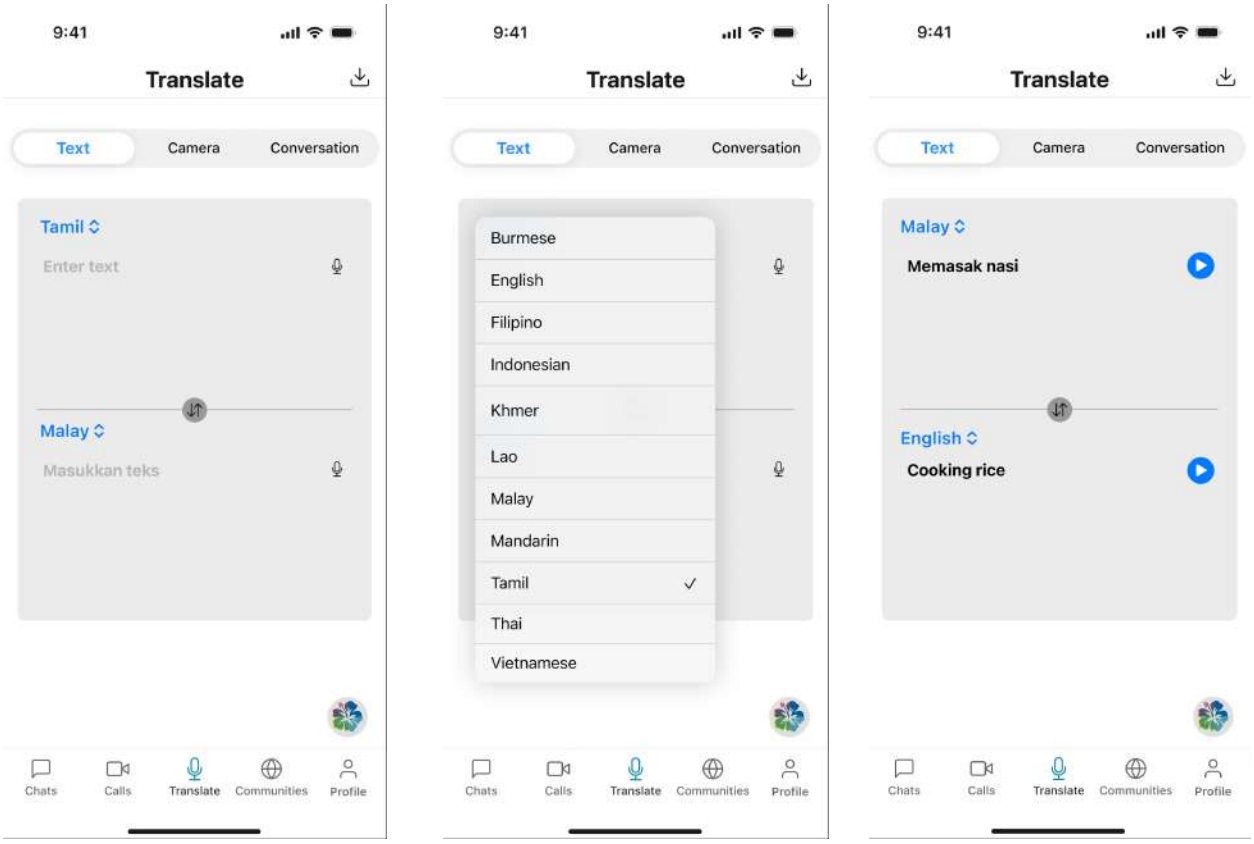
5 Prototype Showcase

5.1 Translate Mode

This mode allows users to translate or communicate with other people in person in ASEAN languages including traditional languages, dialects and sign languages.

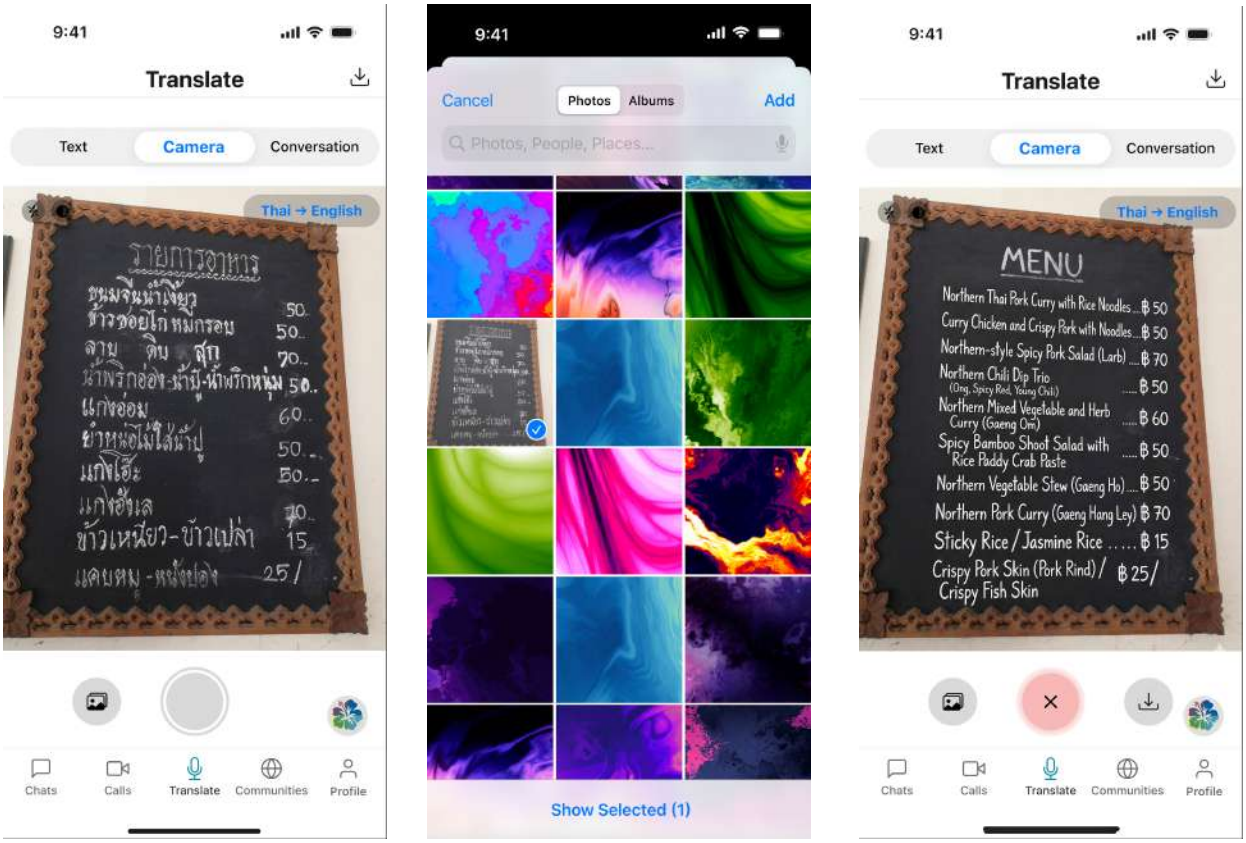
5.1.1 Text Translation

Like any other translator app, users can enter text or use voice to translate. They also can switch between languages.



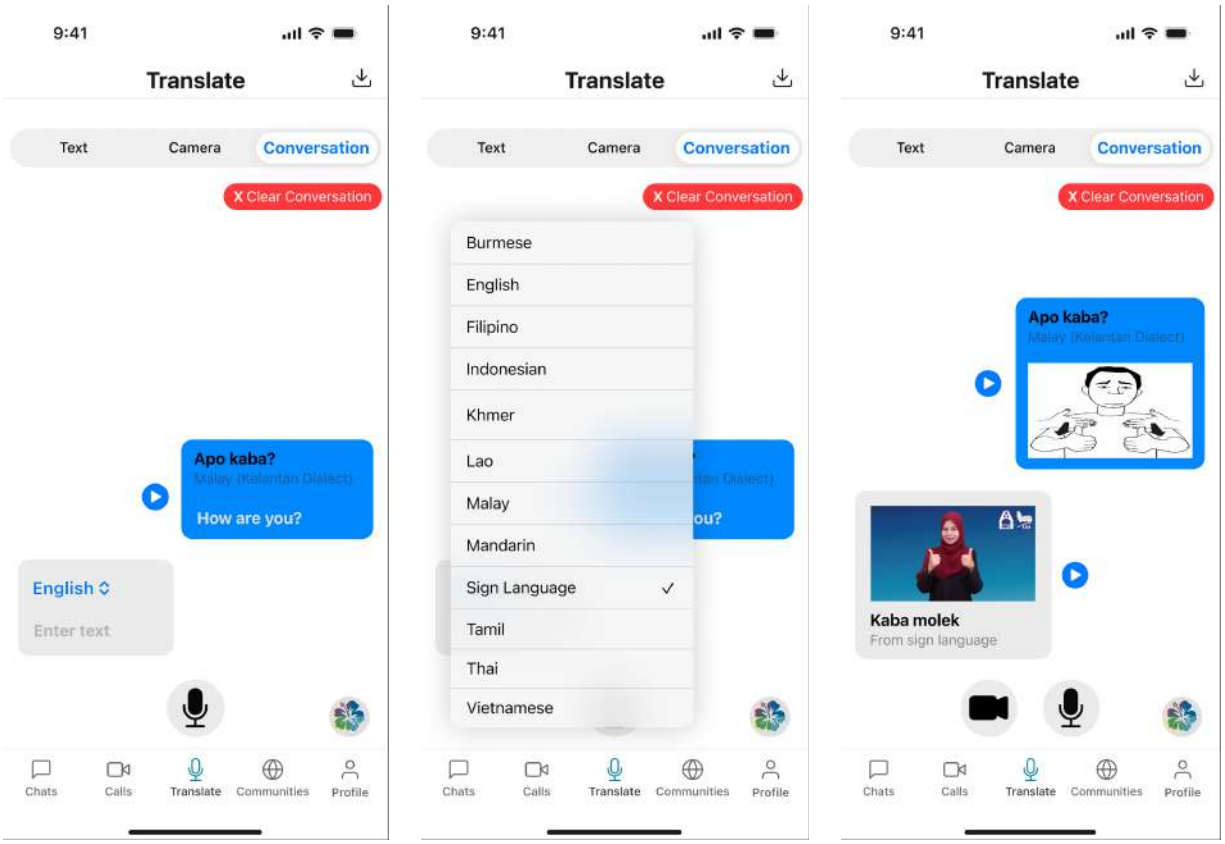
5.1.2 Camera Translation

Users can directly snap or select pictures from their album to translate. This prototype shows that the Thai menu is being translated to English by translating directly from the picture.



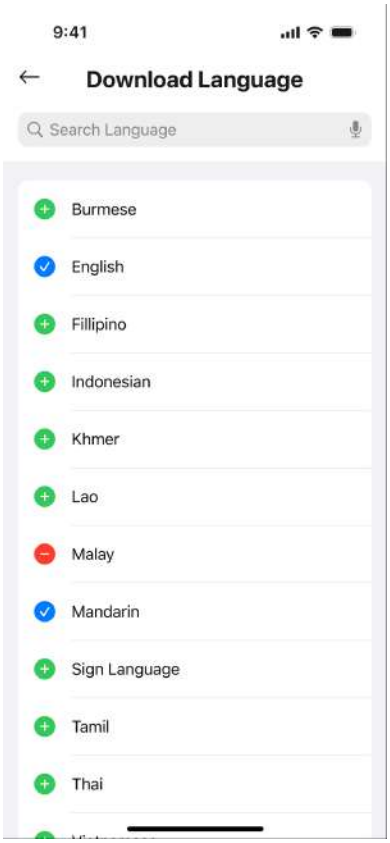
5.1.3 Conversation Translation

Users can either type or speak to have a conversation. The AI will detect the user or speaker language or dialect to translate it accordingly to the user language and dialect.



5.1.4 Download Language

Users can download the languages to be translated especially when traveling offline.

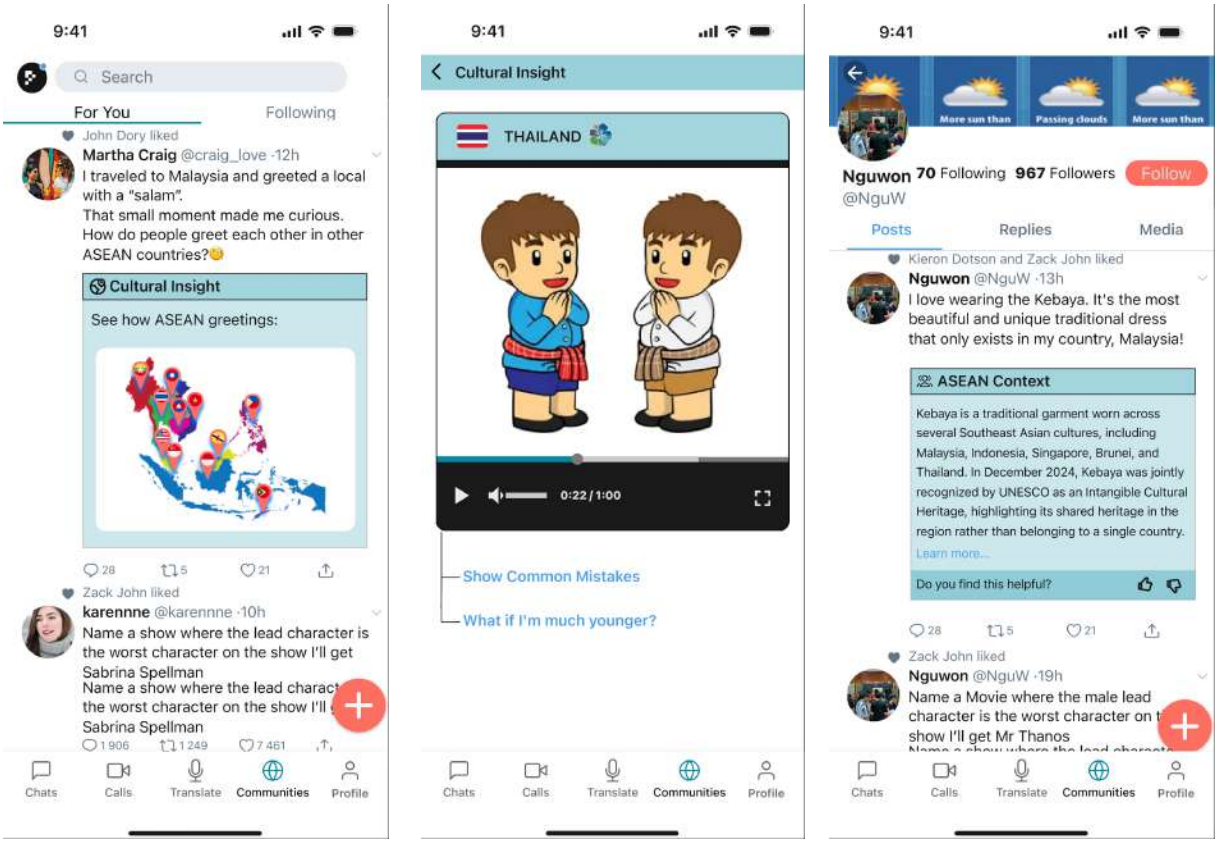


5.2 Community Hub

Community hub is the space where people share their thoughts and cultures while learning about other cultures as well. The AI will help to summarize the content and give some tips and unspoken rules about the culture.

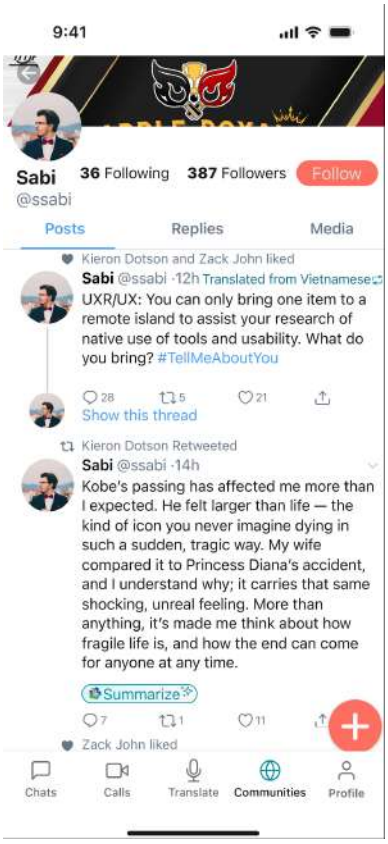
5.2.1 Cultural Insight

AI summarize the post about the culture posted.



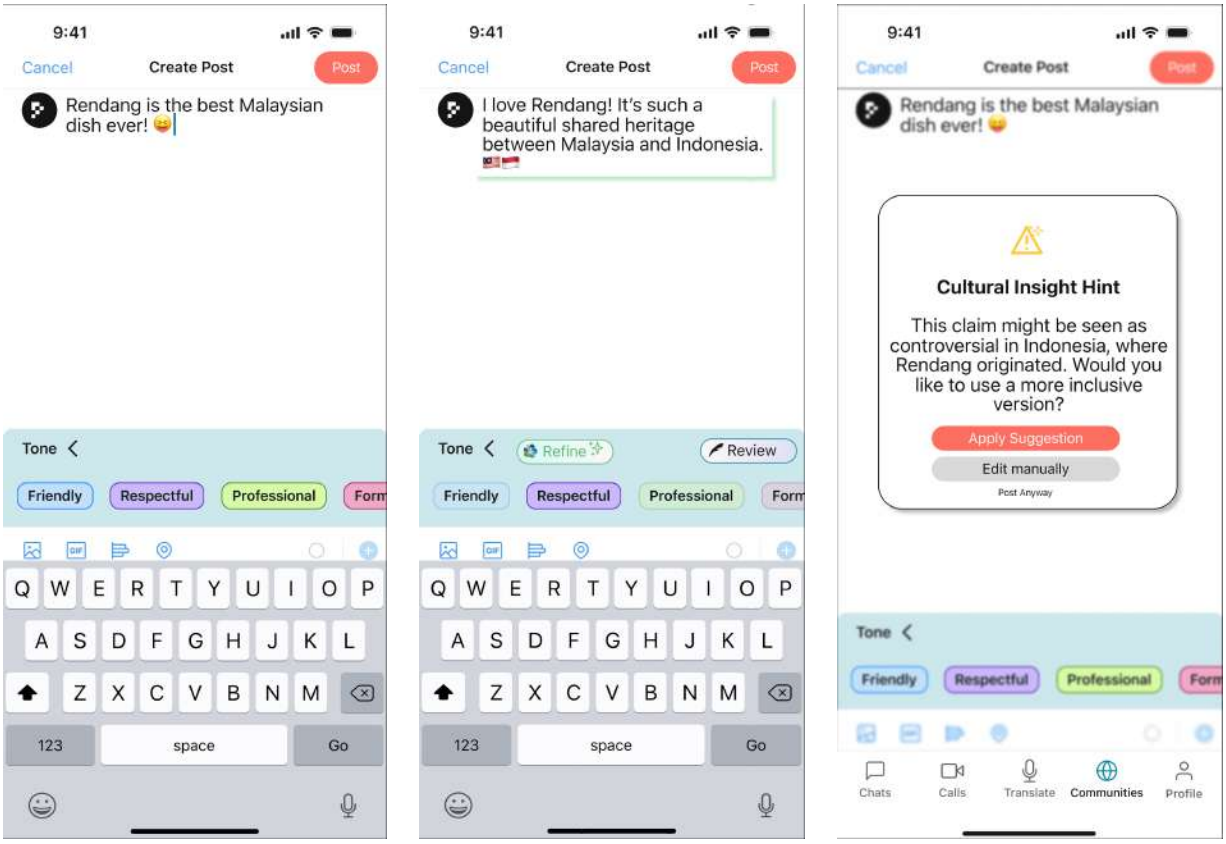
5.2.2 Translate Post

Translate the post from another language into the user preferred language.



5.2.3 Rephrase Post

AI can detect posts that may violate or cause misunderstanding to the other culture and rephrase or correct the tone accordingly.

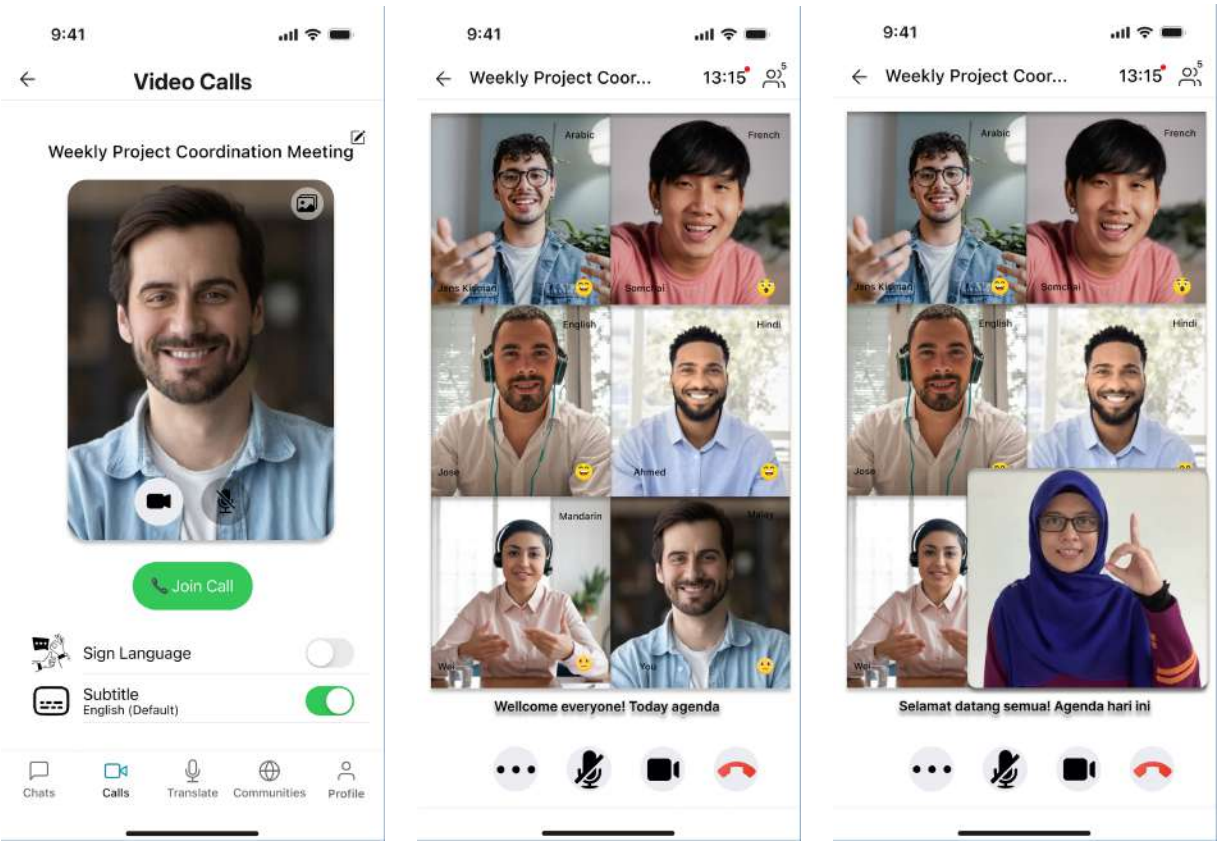


5.3 Collaboration Mode

Collaboration mode allows users especially those who are students or people having meetings with multiple languages by translating into subtitles or sign languages. The AI can also summarize the whole meeting for quick recap and understanding.

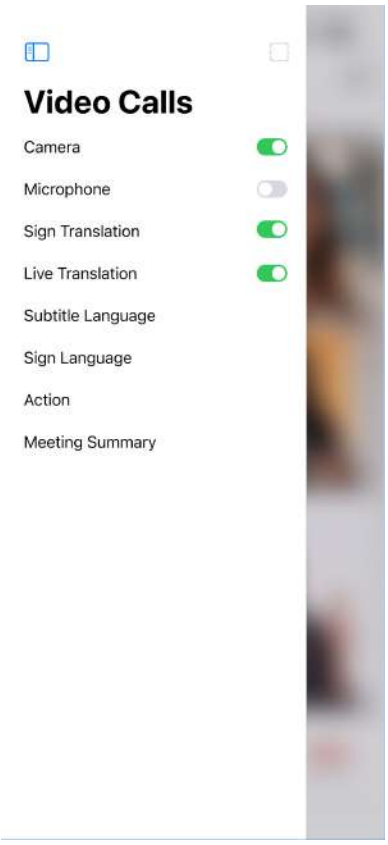
5.3.1 Subtitles and Sign Translations

The AI will translate the whole meeting into subtitles or sign languages while displaying the correct tones and translations through language spoken and face detection. It also can display the other people's mood and language so that users can be aware if they accidentally misunderstood their culture.



5.3.2 Meeting Summary

The AI will summarize all the keynotes and takeaways of the meeting for quick recap and for those who are hearing impaired.



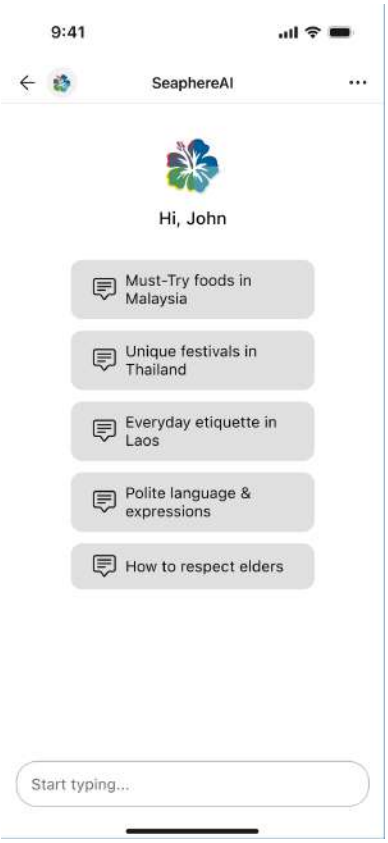
5.4 Chat

5.4.1 Two way translation

Translate your conversation automatically so that the conversation can goes on seamlessly.

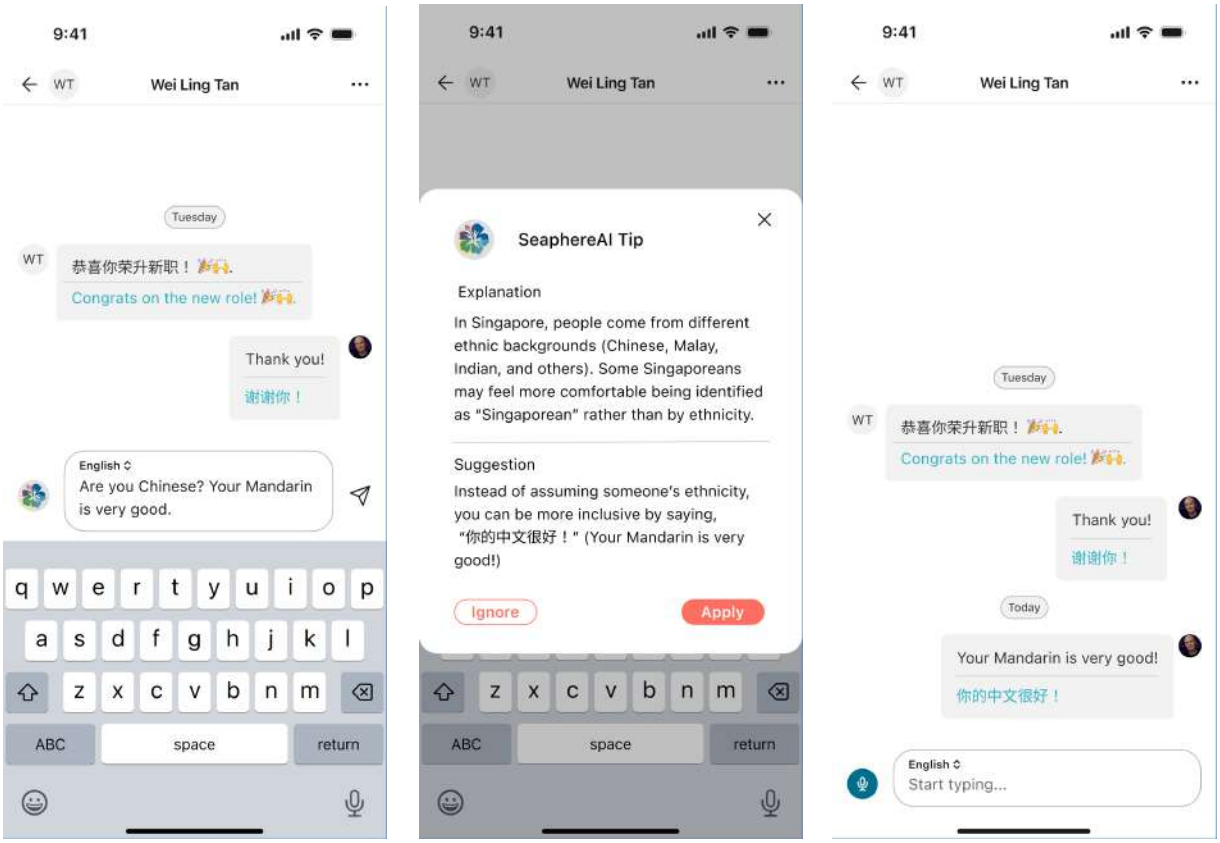
5.4.2 Chatbot

Unlike any Chatbot, Seaphere AI will help and assist the user in understanding culture.



5.4.3 AI Text Refinement

AI will give suggestions based on the text typed by the user to avoid misunderstanding.



6 AI Acknowledgement

Seaphere utilized several AI tools during the ideation, design and documentation phases. These uses of AI tools help to improve productivity, generate ideas, and refine overall quality of prototype and report.

The first Artificial Intelligent (AI) tool used is ChatGPT. ChatGPT was used to assist in ideation phases, it helps during the brainstorming phases in exploring the possibilities features for the Seaphere. In addition, chatGPT also helps in organizing the content of documentation and generating dummy responses inside the prototype. Furthermore, ChatGPT assists in suggesting

various user interface designs that helps enhance user usability and overall user experience of the prototype. Next, Google Gemini being used during the ideation phases and task distribution. This is to ensure that each member of the team is contributing and receives a fair share of work which is also an act to prevent potential disputes. By distributing the task, we manage to execute the project more smoothly and efficiently.

7 Conclusion

In conclusion, ASEAN's cultural diversity presents both an opportunity and challenges for cultural connection among Southeast Asian students. While the region shares historical and economic ties, gaps in cultural understanding and communication styles still exist and can still lead to misunderstandings. The absence of an inclusive communication platform highlights the need for more accessible and culturally aware digital solutions, particularly those that support hearing-impaired and non-literate individuals. By integrating AI, Seaphere aims to address these gaps by providing a communication platform that facilitates interaction across ASEAN. Through its capabilities, students from different linguistic backgrounds can participate in discussions without missing important messages. Additionally, the lack of knowledge about ASEAN cultures and societies can be addressed through SeaphereAI, which provides insights and information to help users better understand the diverse communities within the region.

Looking ahead, Seaphere has the potential to evolve into a broader social and professional ecosystem for ASEAN youth. One proposed enhancement is the introduction of an **"ASEAN Life" Story feature**, inspired by platforms such as WhatsApp and Instagram. This feature would allow students to share short, 24-hour updates about their daily cultural activities, local festivals, and campus life. By sharing these everyday experiences, students from different countries can gain a deeper and more personal understanding of life across Southeast Asia.

Furthermore, the platform aims to bridge the gap between academic learning and regional career opportunities by introducing the **Seaphere Professional ASEAN Certification**. Inspired by programs such as the Google Career Certificates, this initiative would provide AI-guided courses on topics such as regional business etiquette, cross-border collaboration, and ASEAN socio-economic development. Upon completion, participants would receive a verified digital certificate that can be shared on professional platforms like LinkedIn. By offering a recognized credential across the region, this feature could help improve students' employability within the ASEAN workforce while also strengthening their cultural awareness and collaborative skills.

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